Algorithms

FLOW – Part 1

Last Week

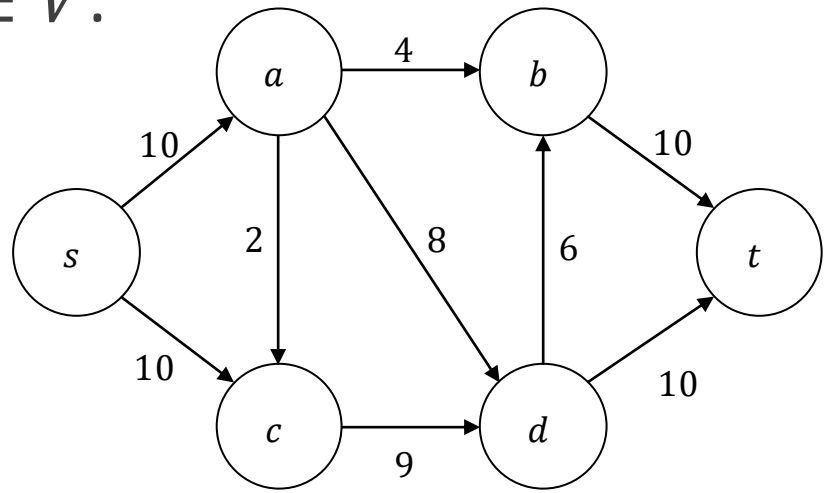
- Minimum Spanning Trees (MST)
  - Prim Algorithm

This Week

- Flow networks
- Maximum flow

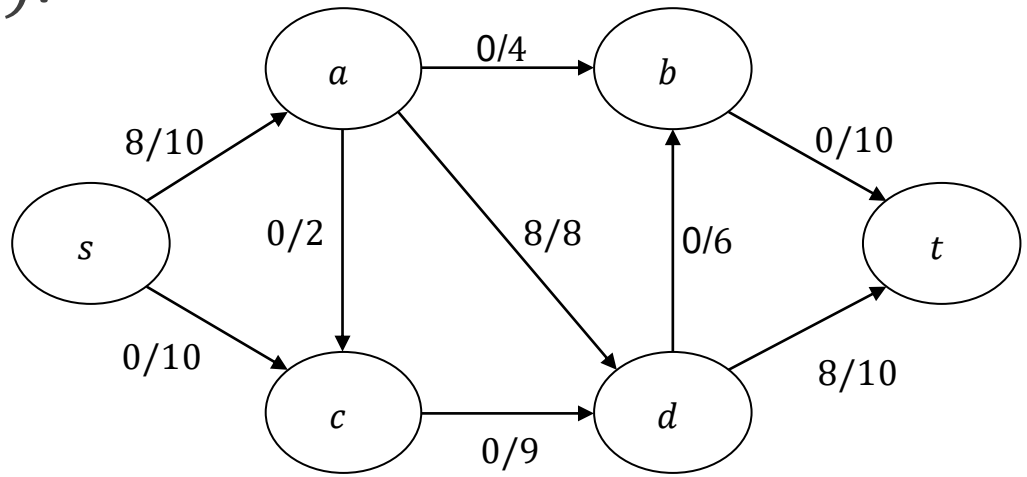
Flow

- The input to a flow problem consists of:
  1. a directed graph (network) $G = (V, E)$
  2. a capacity function on the edges $c: E \to R$
  3. source and target vertices $s, t \in V$.



Flow

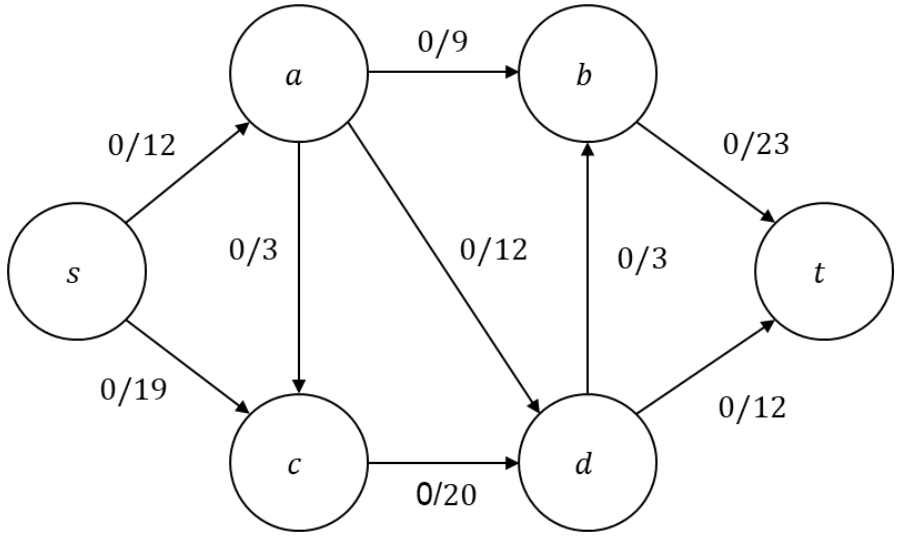
- Definition 1. A flow function $f: E \to R$ is a function on the edges such that:
  - Edge Condition: For all $(u, v) \in E, 0 \leq f(u, v) \leq c(u, v)$.
  - Vertex Condition: For any fixed $v \in V - \{s, t\}$, $\sum_{(u,v)\in E} f(u, v) = \sum_{(v,w)\in E} f(v, w)$.



Flow

- Definition 2. The flow value is defined as:

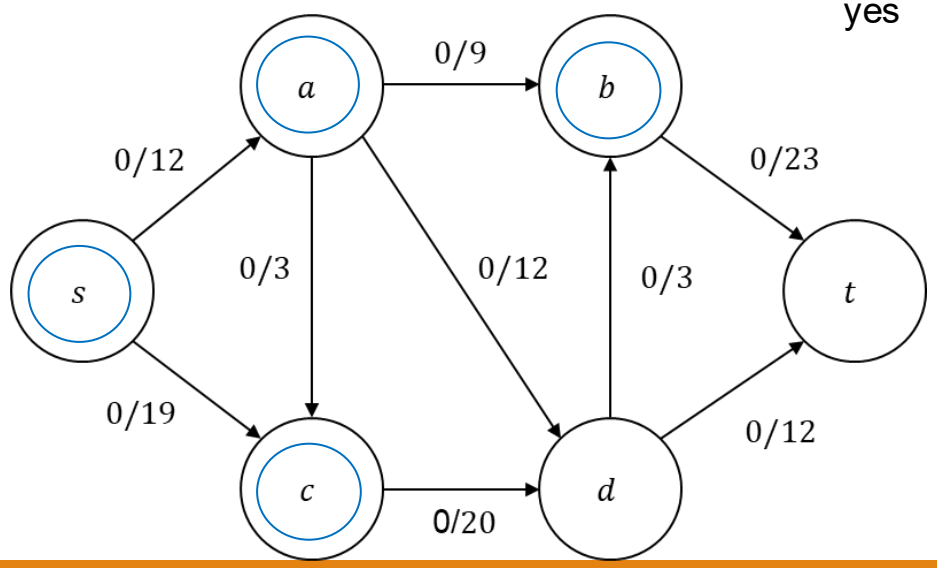
$$\mathrm{F} = \sum_{(s,v)\in E} f(s,v) - \sum_{(u,s)\in E} f(u,s)$$



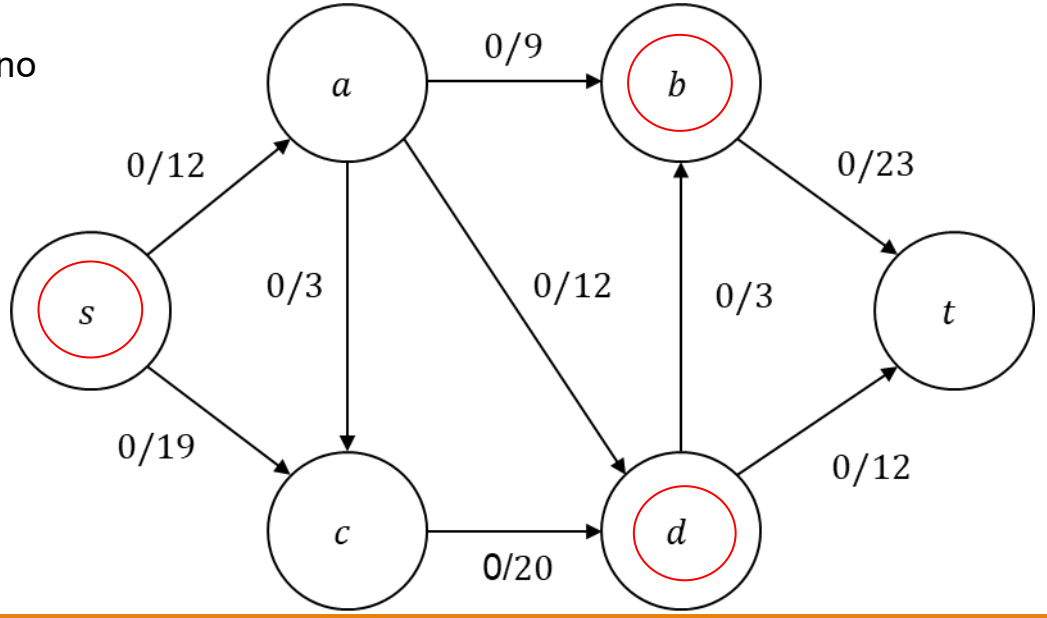
Flow

- Definition 3. An s-t cut $(S, V \backslash S)$ is a partition of the vertices where the source $s \in S$ and the sink $t \notin S$ are not in the same side of the partition. The vertices in S are all reachable from s.

yes



no



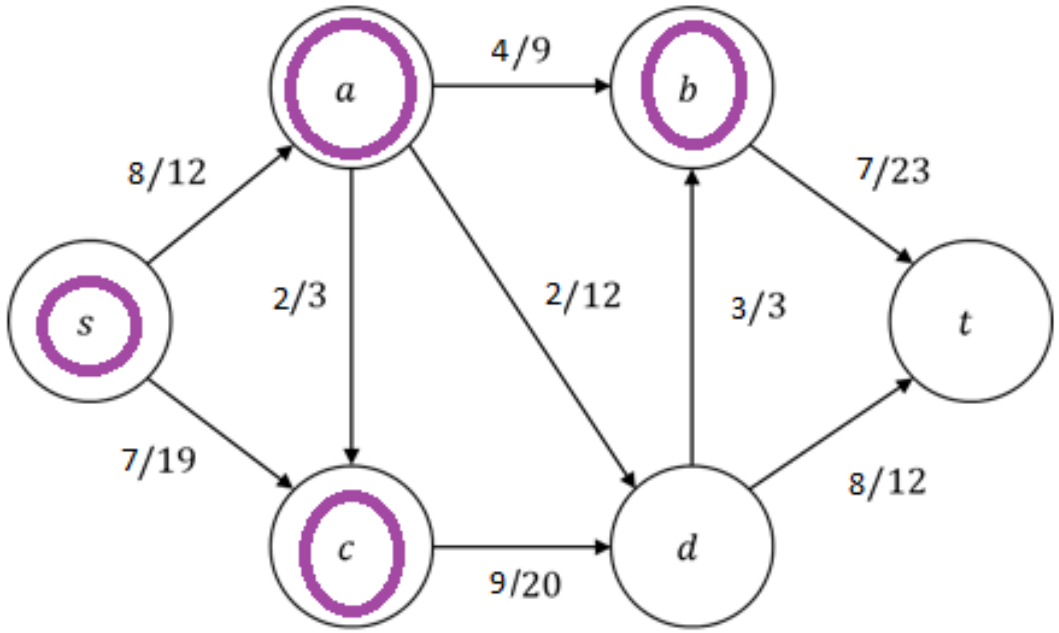
Flow

Claim. For any cut S,

$$\mathrm{F} = \sum_{e \in out(S)} f(e) - \sum_{e \in in(S)} f(e)$$

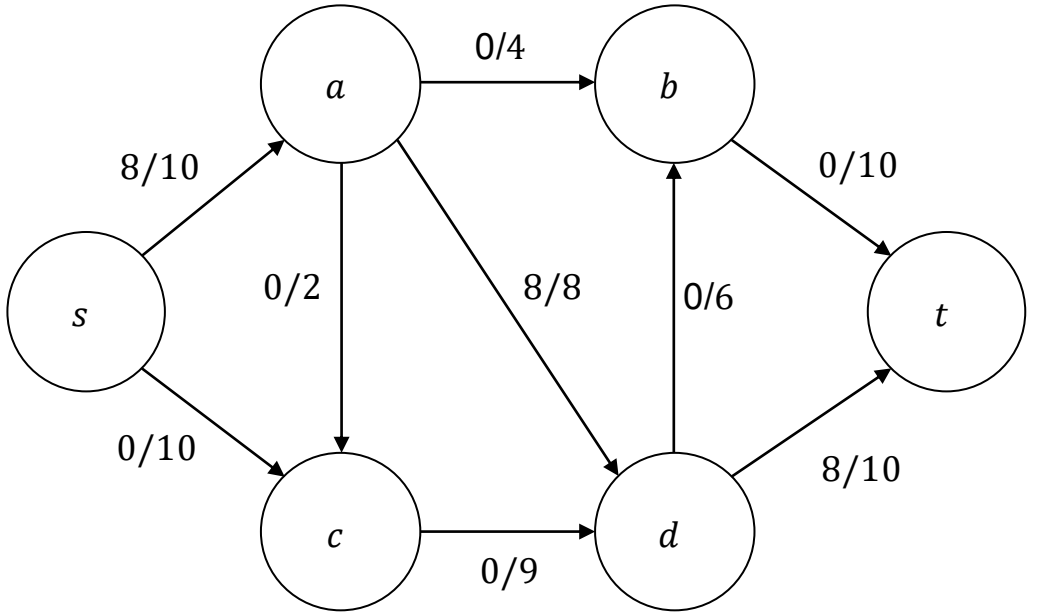
$f_{out}(S) = 7 + 2 + 9 = 18$
$f_{in}(S) = 3$
F = 8+7 = 15 (out of s)
F = 18-3 = 15

The capacity of a cut is the total capacity of edges in out(S).
C(S)= 23+12+20



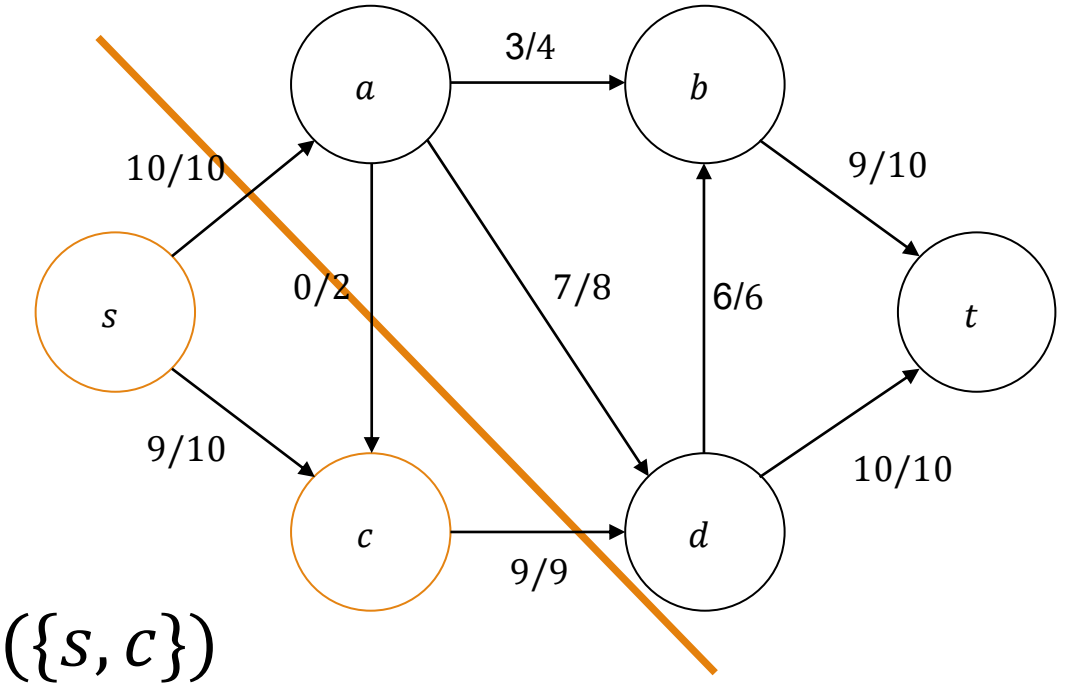
Flow Network

- Legal flow on G:



Flow Network – MinCut Theorem

- The maximum flow in the network is equal to the minimum capacity of an s-t cut



$F = 19$

$C(\{s, c\}) = 19$

$= \ capacity\ on\ out(\{s, c\})$

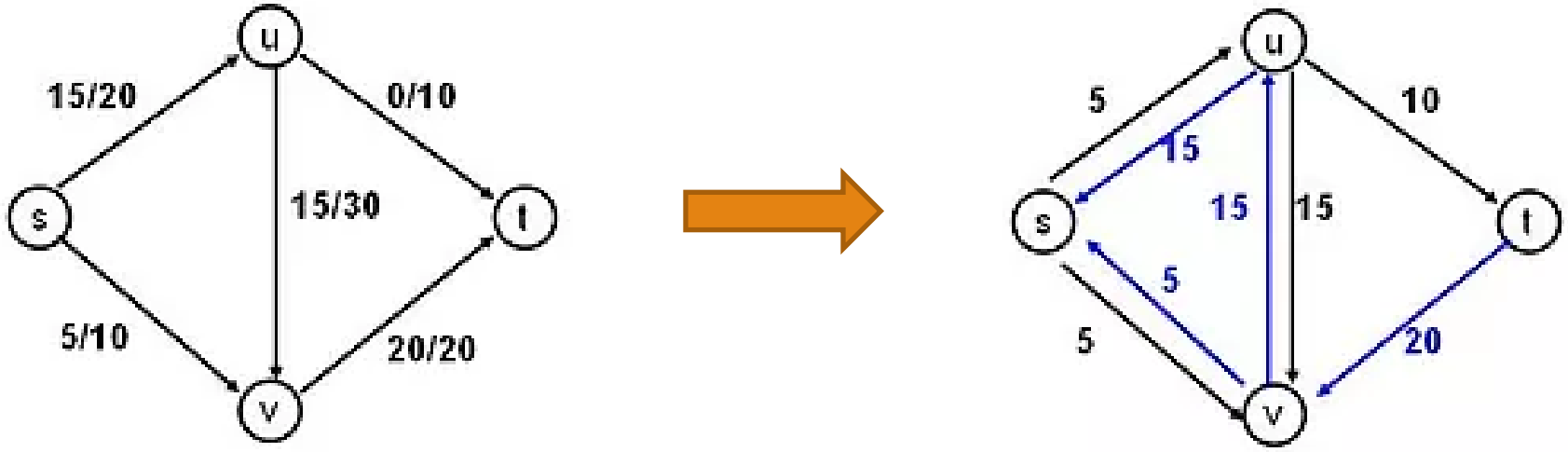
The Residual Graph

- The residual graph with respect to a given flow f is given by $R_f = (V, E')$, where:

$$E' = \{(u,v) \in E : f(u,v) < c(u,v)\} \cup \{(v,u) | \forall (u,v) \in E \; s.t. \; f(u,v) > 0\}$$

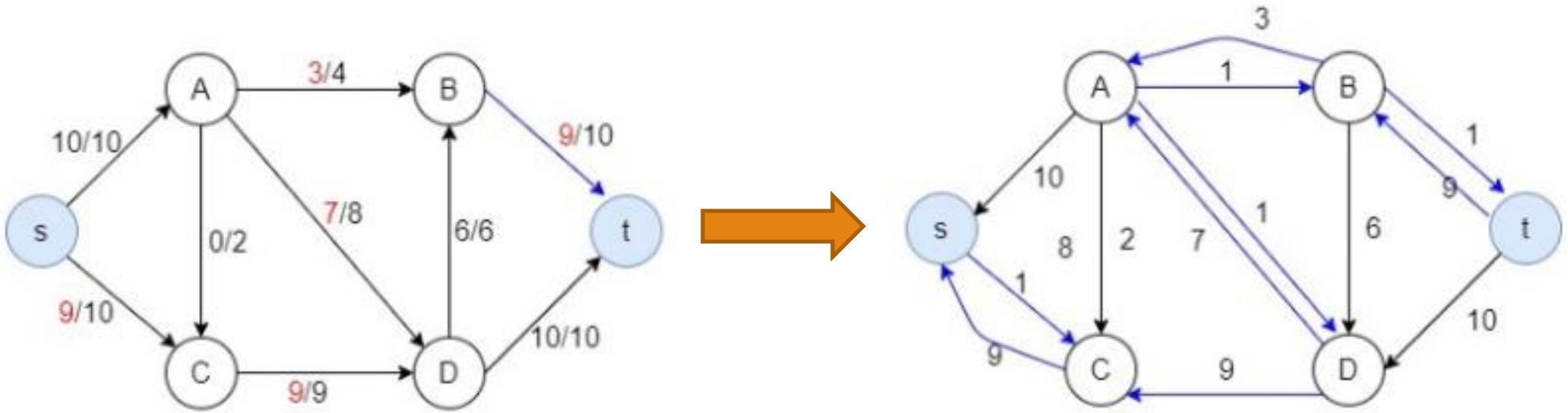
- The capacity of a forward edge (u,v) in R_f is given by $c'(u,v) = c(u,v) - f(u,v)$ and the capacity of a backward edge is given by $c'(v,u) = f(u,v)$.

The Residual Graph



Claim

- f is a max flow $\Leftrightarrow$ there is no path from s to t (augmenting path) in the residual graph R_f



Ford Fulkerson

Algorithm 1 FORD-FULKERSON(G, c)

```
1: Initialize flow on all edges to 0
2: while there is a path p from s to t in R do
3:     δ = min_{e∈p} c'(e)
4:     Flow δ along p and update forward and backward edges accordingly
5:     Update R along p
```

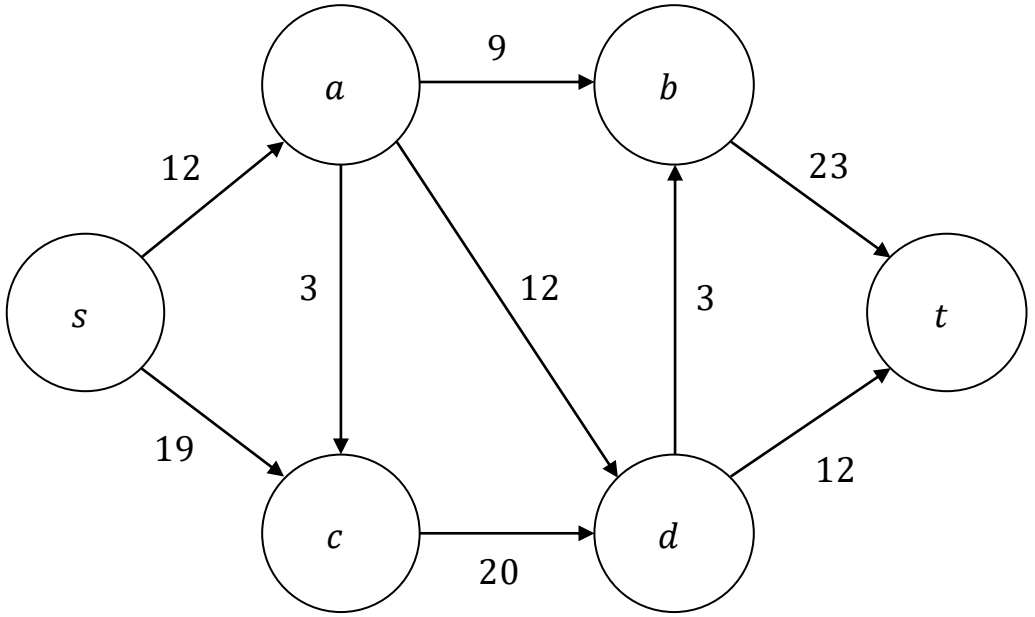
Ford Fulkerson

Time Complexity:

- Each iteration takes $O(m)$.
- If we assume the capacity values are integers, then there are at most f^* iterations, Where f^* is the value of the max flow.
- Total complexity: $O(m \cdot f^*)$.
- If in every iteration p is a shortest $s-t$ path (Edmond-Karp), then there are at most $O(mn)$ iterations, implying total time complexity $O(m^2 n)$.

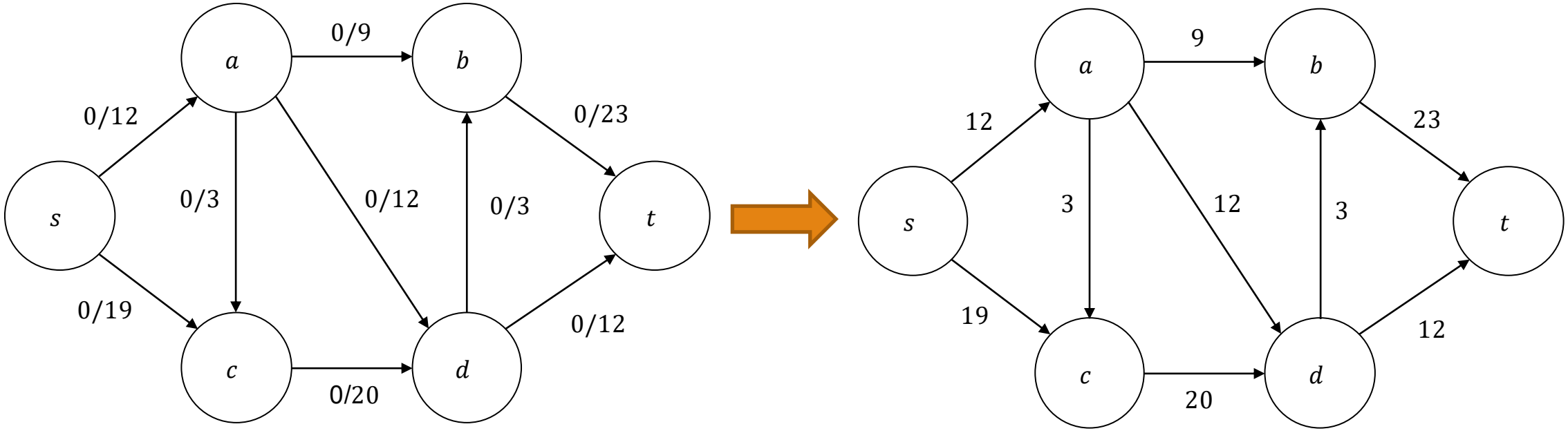
Exercise 1 - Ford Fulkerson

Find a maximum flow in the following graph using FF:



Exercise 1 - Solution

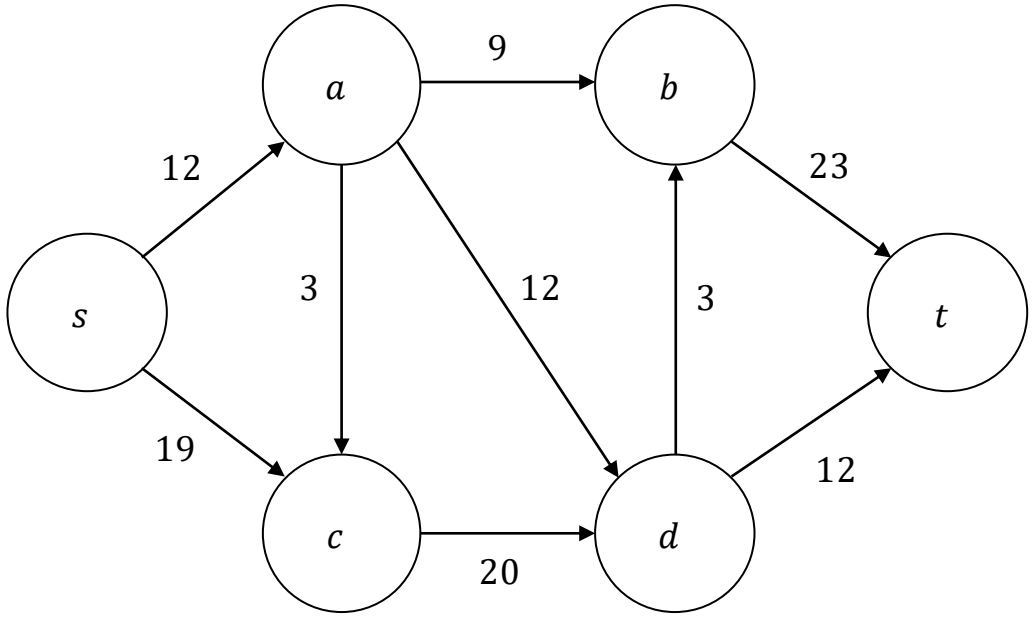
Start with $f = 0$ and build R_f:

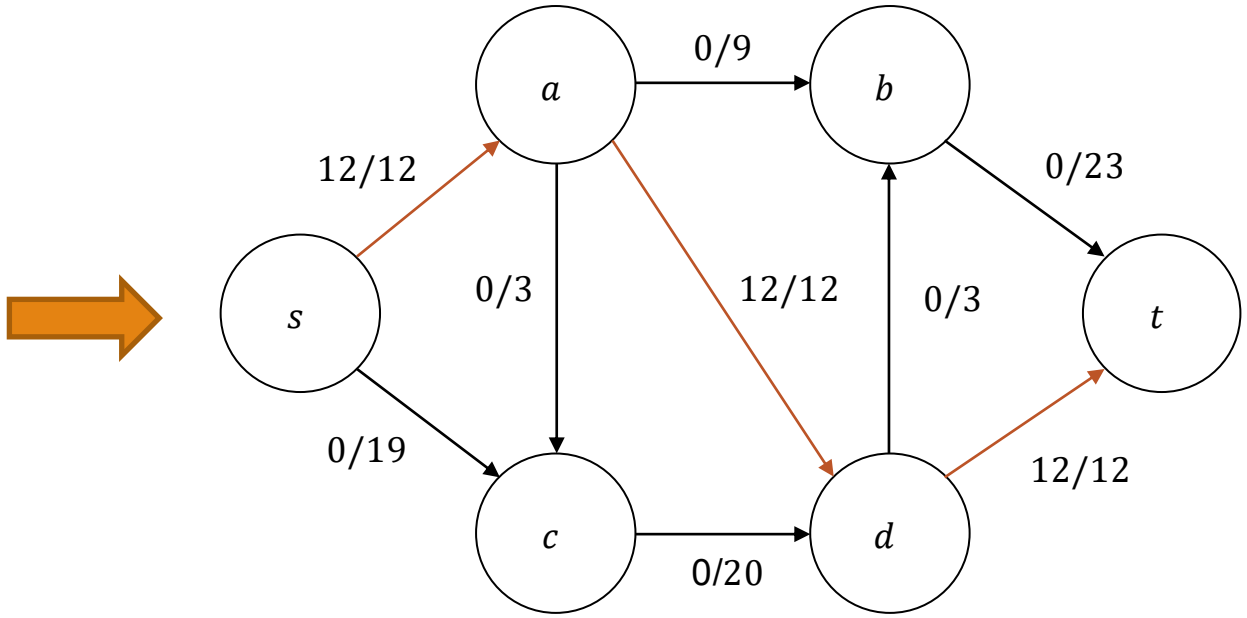


$F = 0$

Exercise 1 - Solution

Choose a path: $s \to a \to d \to t$ and update f:



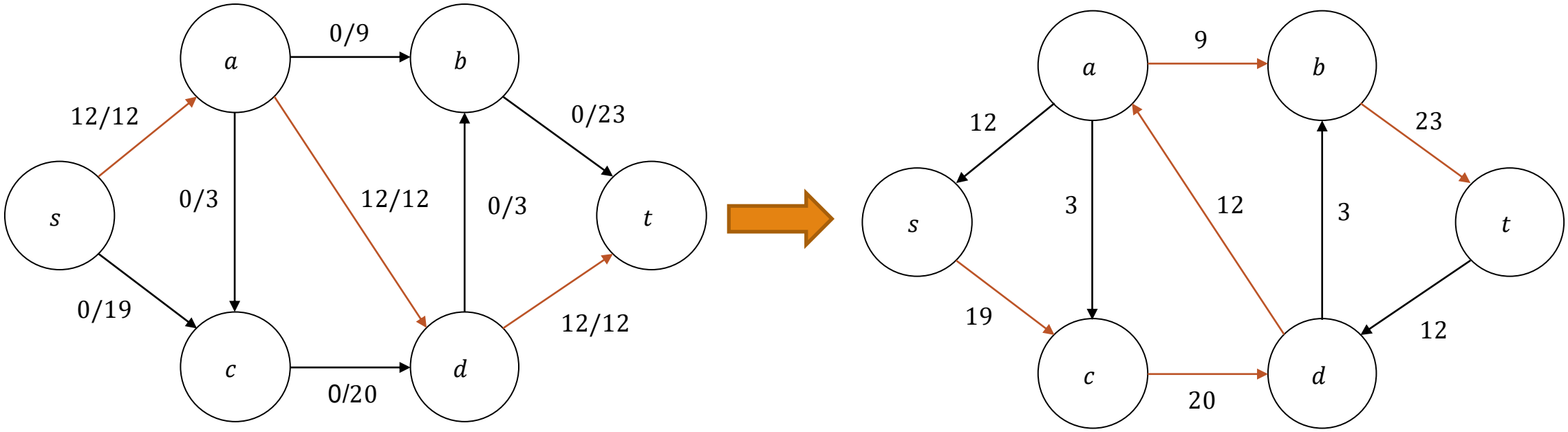


$F = 12$

$\delta = 12$

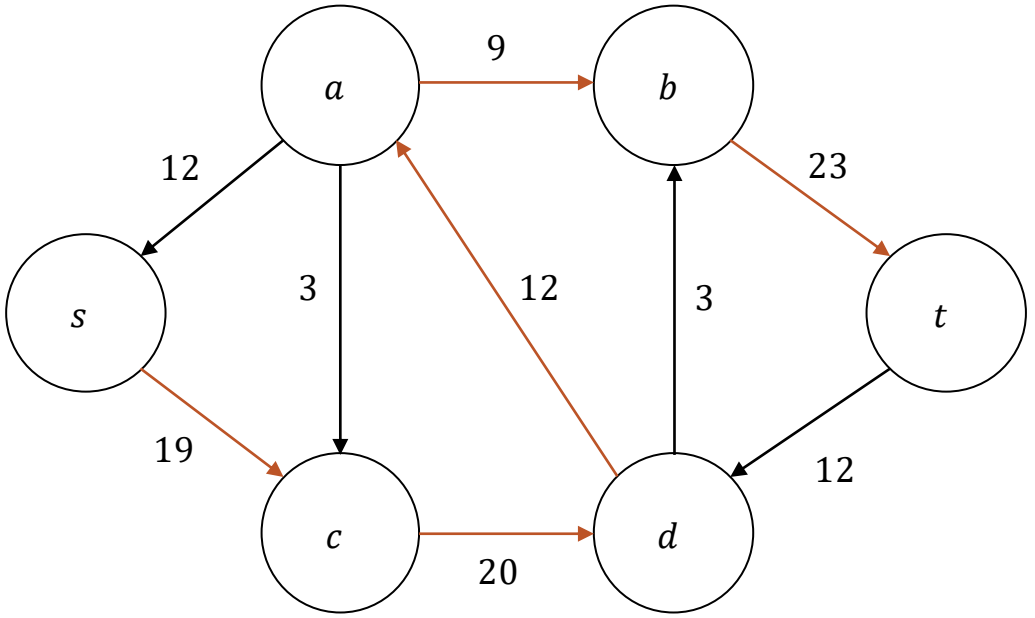
Exercise 1 - Solution

Build R_f:

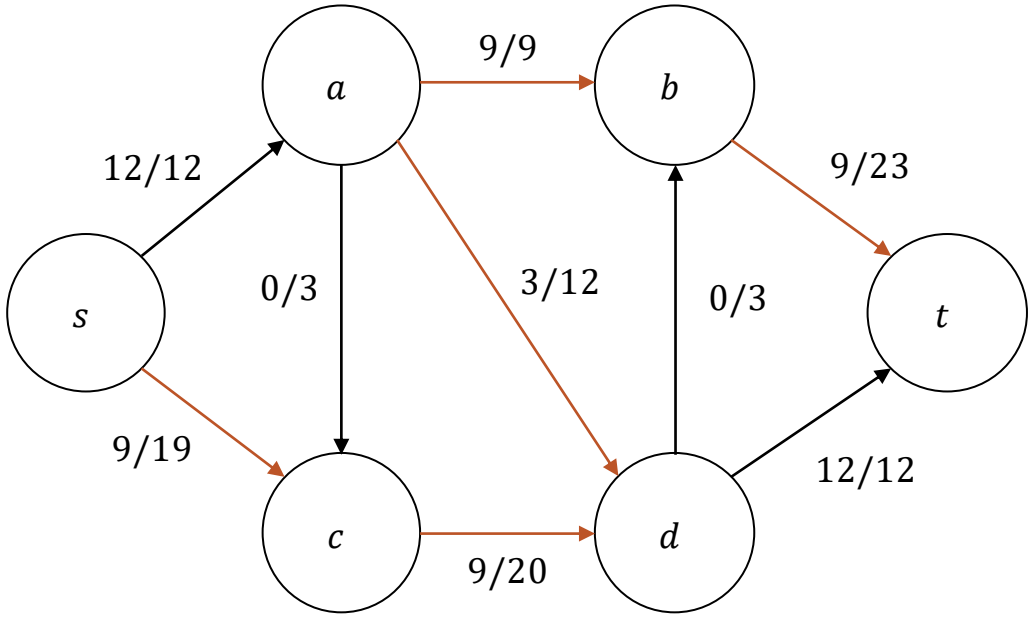


Exercise 1 - Solution

Choose a path: $s \to c \to d \to a \to b \to t$ and update f:





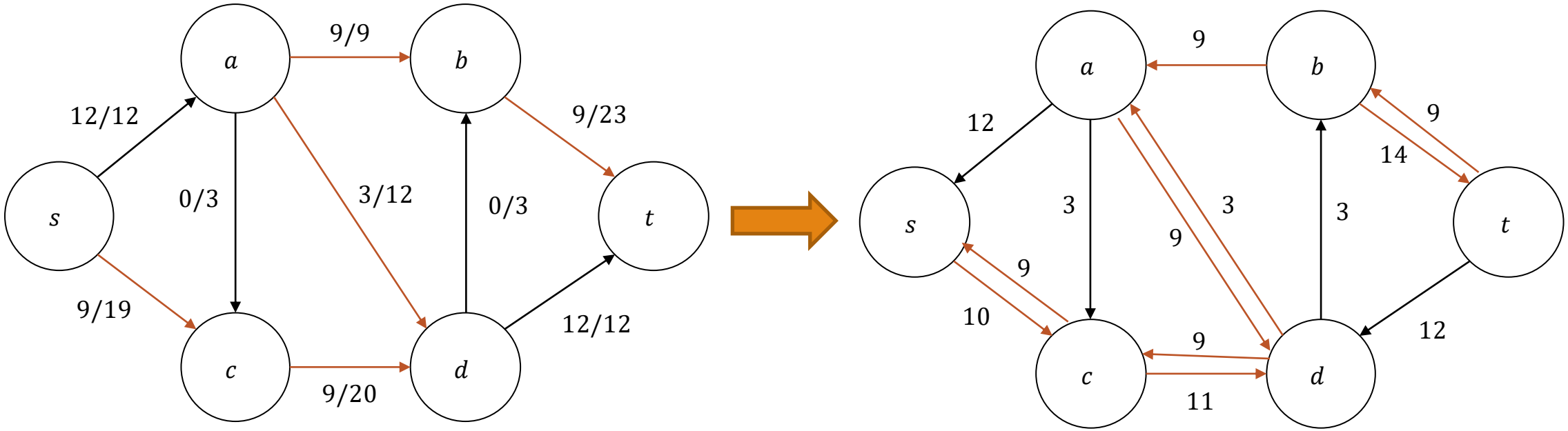


$F = 21$

$\delta = 9$

Exercise 1 - Solution

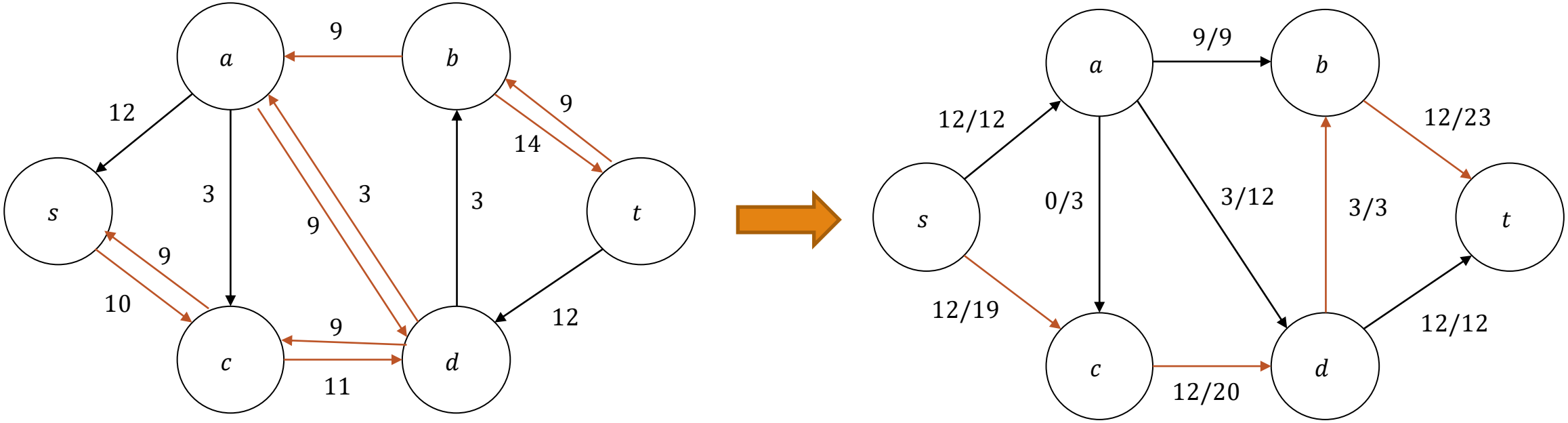
Build R_f:



$F = 21$

Exercise 1 - Solution

Choose a path: $s \to c \to d \to b \to t$ and update f:

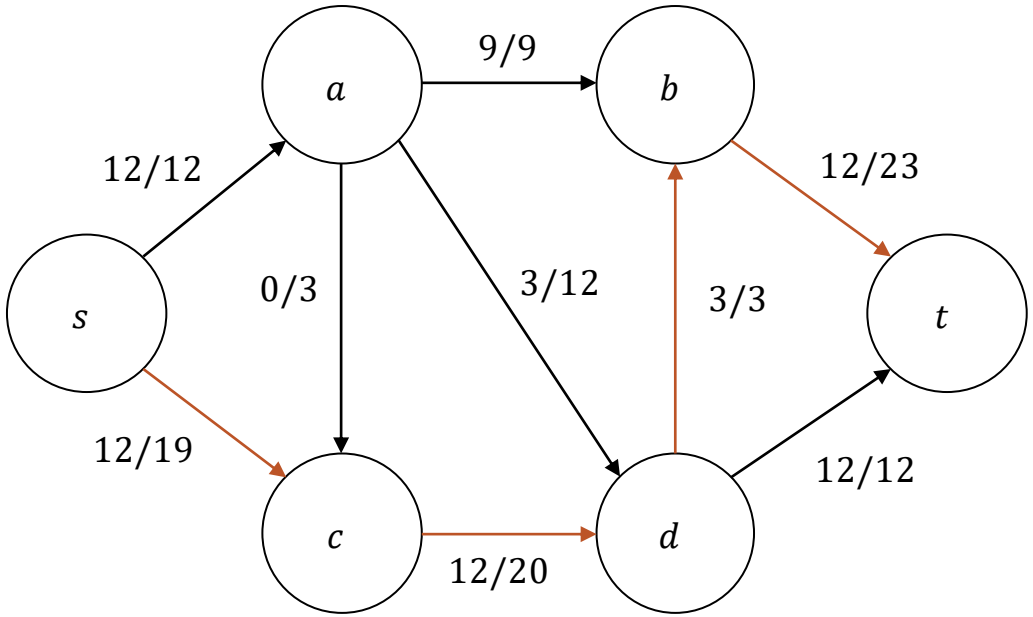


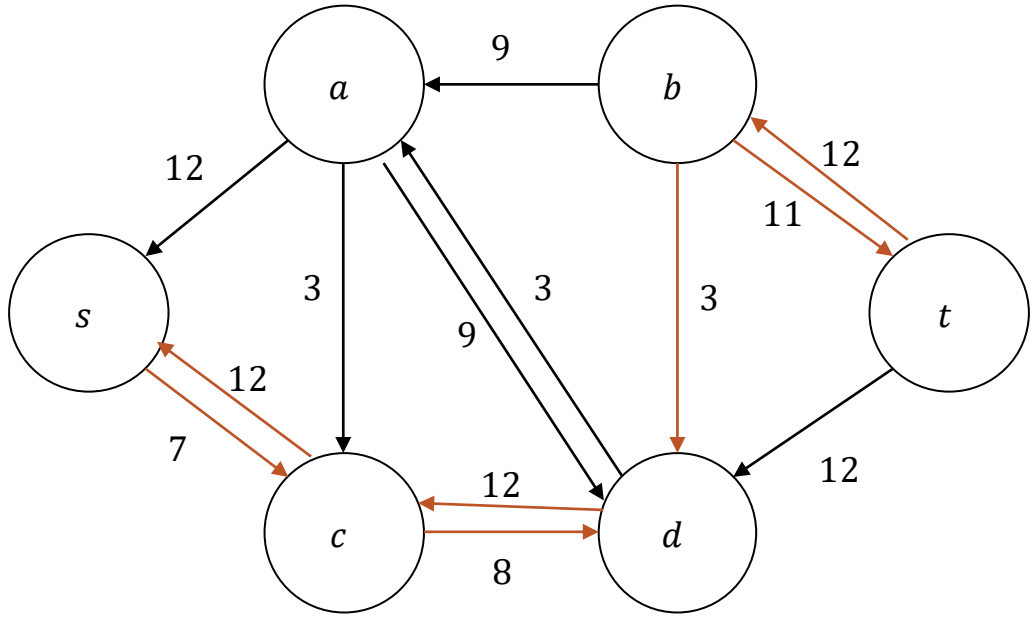
$F = 24$

$\delta = 3$

Exercise 1 - Ford Fulkerson

Build R_f:



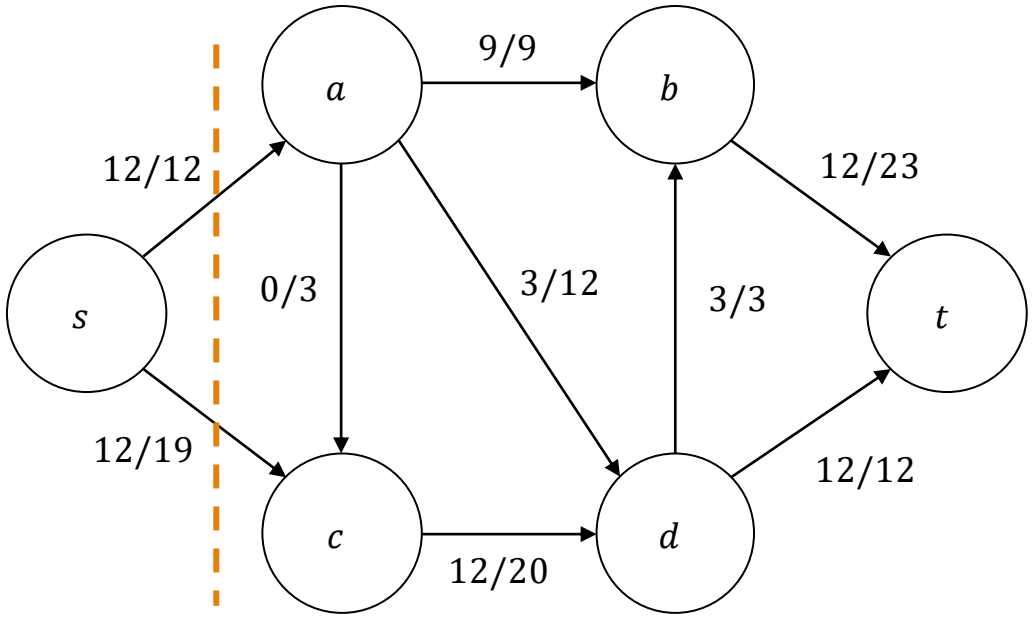


$F = 24$

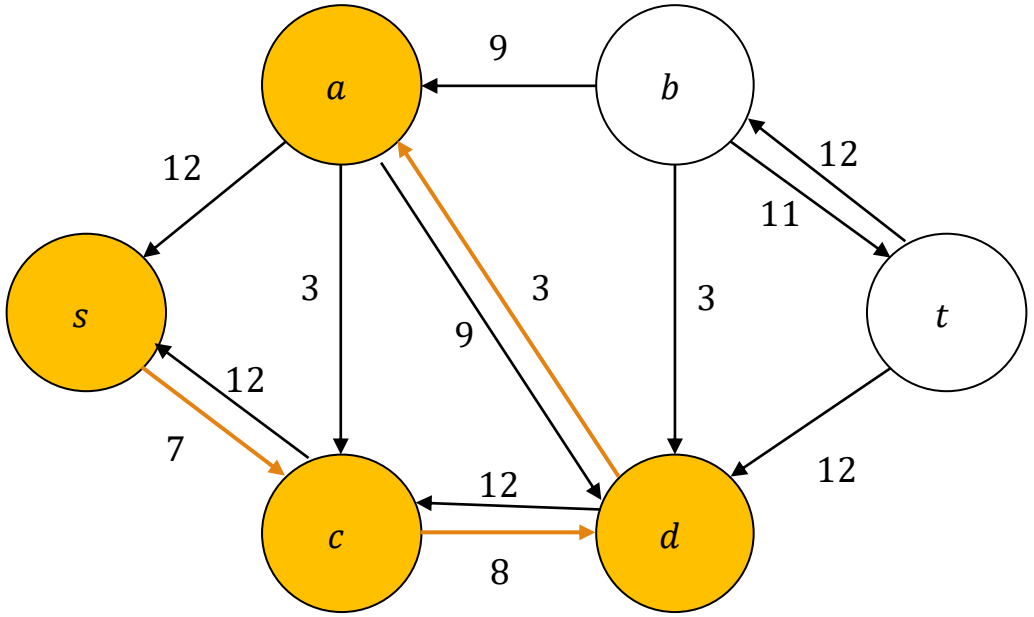
There is no path from s to t!

Exercise 1 - Ford Fulkerson

$$F = f^* = 24$$



Min cut: $S = \{s, a, c, d\}$



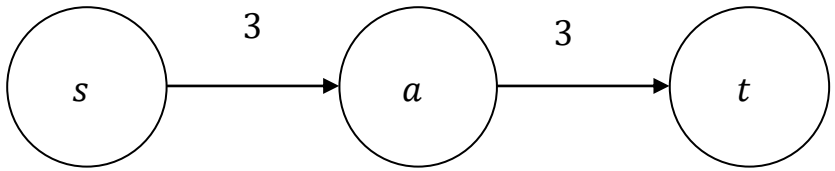
Note that all edges in Out(S) are saturated

Exercise 2

- An edge is upward critical if increasing the capacity of this edge increases the maximum flow value.
- Does every flow network contain an upward critical edge?
- Suggest an algorithm that finds all upward critical edges in a given graph.

Exercise 2 - Solution

- First, we notice that not every flow network has an upwards critical edge.
- For example, a simple path network with 2 edges having equal capacities.



Exercise 2 – Solution 1

Naive solution:

1. Perform FF on G and find $f^*(G)$.
2. For every $e \in E$:
   2.1. let G'_e be a network produced from G by increasing $c(e)$ by 1.
   2.2. Run FF and output e if $f^*(G'_e) > f^*(G)$.

Time: $O(|E|^3|V|)$

Exercise 2 – Solution 2

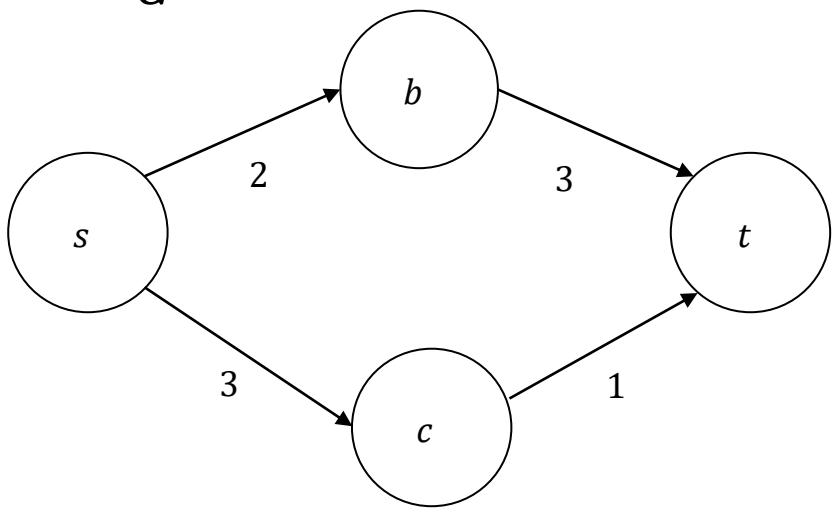
Algorithm:

1. Find the maximum flow f^* using EK.
2. Let R_f be the last residual network. There is no path from s to t in R_f.
3. Run BFS from s on R_f and find all the vertices reachable from s. Denote this set by S.
4. Run BFS from t on R_f with reversed edges to find all vertices from which there's a path to t. Denote this set by T.
5. Every edge $(u, v) \in E$ such that $u \in S, v \in T$ is upward critical.

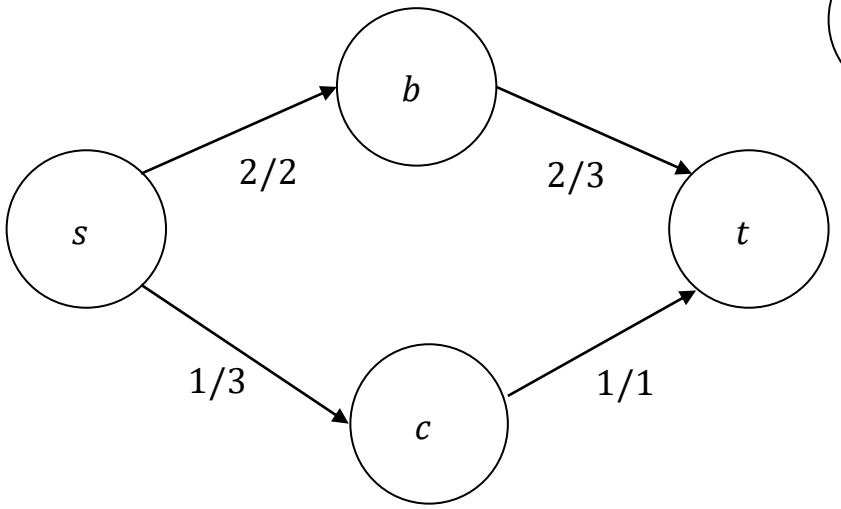
Exercise 2 – Solution 2

- Example: $(s, b), (c, t)$ are upwards critical

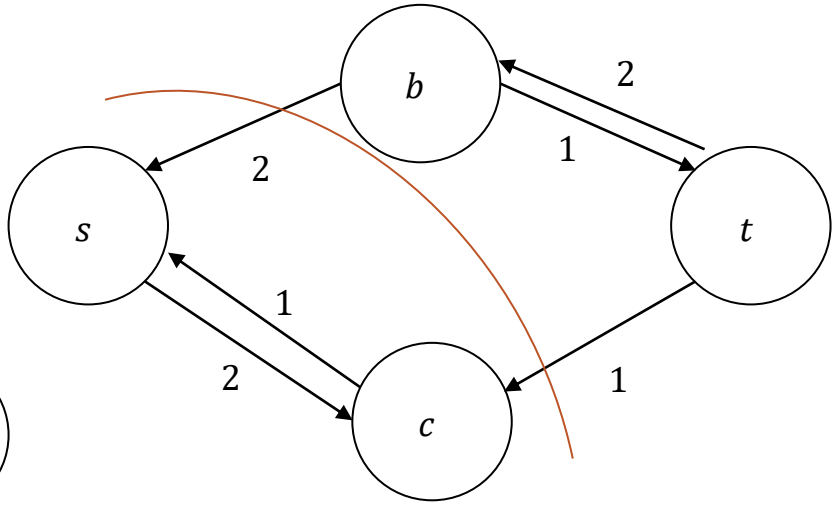
G



Max flow



R_f



$S = \{s, c\}$
$T = \{b, t\}$

Exercise 2 – Correctness

- Increasing the capacity of any edge (u, v), s.t. $u \in S$, $v \in T$ results in an augmenting path, so it is upward-critical.
- Furthermore, every upward-critical edge has this property, so the set of edges found by the algorithm is equal to the set of upward-critical edges.
- Note: An edge is upward critical if it belongs to all min-cuts.

Exercise 2 – Complexity

- The running time of the algorithm is dominated by the calculation of the maximum flow.
- Finding max flow: $O(m^2 n)$ using EK.
- Calculating the sets S and T: $O(m + n)$.
- Finding upward-critical edges: $O(m)$.

All together: $O(m^2 n)$.

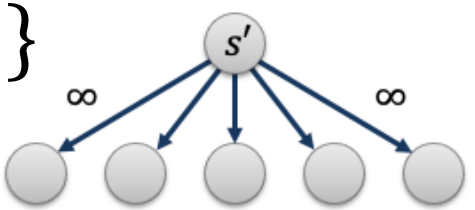
Exercise 3

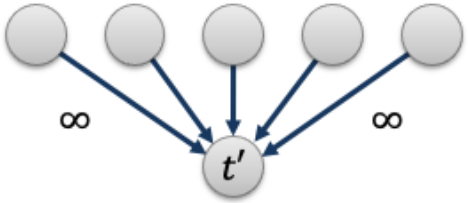
- Let $G = (V, E)$ be a flow network with $a \geq 1$ source vertices $\{s_1 \dots s_a\}$ and $b \geq 1$ target vertices $\{t_1 \dots t_b\}$.
- Flow can be routed from every source to every target.
- The value of a given flow is the total flow out of all sources.
- Suggest an algorithm that finds maximum flow in G.

Exercise 3 - Solution

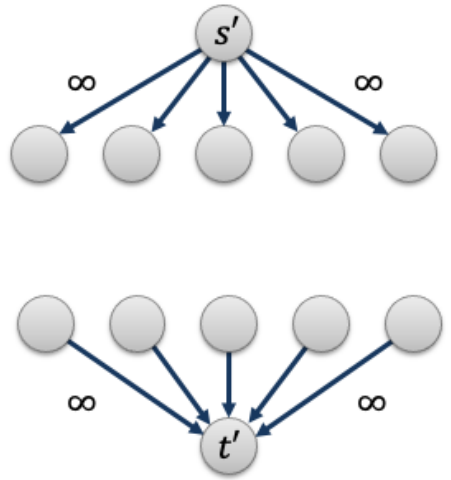
Algorithm:

1. Construct a new graph: $G' = (V', E')$, where:
- $V' = V \cup \{s', t'\}$
- $E' = E \cup \{(s', s_i), (t_j, t') \; \forall 1 \leq i \leq a, 1 \leq j \leq b\}$
- New edges have capacity ∞.
2. Find a max flow from s' to t' in G' .





Exercise 3 - Solution



Correctness:

- Main claim: There is a flow of value k in G' ⟺ there is a flow of value k in G with multiple sources.
- Since the only edges that have infinite capacity are the new edges, every cut that includes all sources on one side and all targets on the other, have finite capacity.
- Thus, the value of the flow in the network is bounded by the value of the min cut that separates all sources from all targets as required.